MODEL LAB · 2026-10-14

Every prior, same slate

The held-out scores that decided which prior is served, then how each one would have ranked today's hitters.

Served (full season)
served prior

11
priors compared

27,888
held-out plate appearances

3.04%
held-out HR rate

0.5974
best AUC · KNN comps

Held-out scores

Fitted on plate appearances before 2026-09-02, scored on the 27,888 that came after (actual HR rate 3.04%). No outcome in the test window was visible during fitting. **Read AUC first** — the slate ranks hitters and takes the top six, so ordering is what matters. Brier and log loss are dominated by the ~3% base rate and barely separate the priors. Metrics are per plate appearance; the probabilities lower down are per game.

PRIOR	AUC	TOP-DECILE LIFT	BRIER	LOG LOSS	CAL. RMSE
Flat league League HR rate as the prior for every hitter.	0.5744	1.58×	0.02943	0.13530	0.00505
KNN comps Prior is the pooled rate of the nearest comparables.	0.5974	1.56×	0.02939	0.13431	0.00513
SVR Support-vector regression on the same feature matrix.	0.5901	1.50×	0.02944	0.13515	0.00667
Random forest Forest regression; a learned similarity metric.	0.5923	1.51×	0.02942	0.13464	0.00598
Core (KNN+SVR) Blend of KNN and SVR, no forest.	0.5974	1.56×	0.02939	0.13431	0.00513
Forest blend KNN + SVR + forest, weights fitted out-of-fold.	0.5966	1.51×	0.02939	0.13431	0.00544
Linear regression Ordinary least squares on the same features.	0.5917	1.50×	0.02943	0.13478	0.00562
Logistic regression Binomial logistic on the per-PA home run outcome.	0.5934	1.56×	0.02944	0.13481	0.00699
Neural net (torch) Two-layer PyTorch MLP, PA-weighted, trained on the same features.	0.5919	1.58×	0.02941	0.13464	0.00602
XGBoost Gradient-boosted trees, depth 3 with subsampling and L2.	0.5924	1.47×	0.02945	0.13487	0.00700

BLEND WEIGHTS (FOREST VARIANT)

KNN **0.70** SVR **0.00** RF **0.30**

SHRINKAGE K PER PRIOR

league **255** knn **255** svr **255** rf **255** core **255** ensemble **255** linear **255** logistic **255** nn **255** xgb **255**

Estimated by method of moments on out-of-fold residuals, not hand-tuned.

Thin-sample subset

The same held-out window, restricted to hitters with fewer than 100 plate appearances before the cutoff (3,279 PA, actual HR rate 2.29%). For a regular with 600 PA the observed record swamps any prior, so the priors mostly separate here.

PRIOR	AUC	TOP-DECILE LIFT	BRIER	LOG LOSS	CAL. RMSE
Flat league League HR rate as the prior for every hitter.	0.5776	1.33×	0.02238	0.10950	0.01207
KNN comps Prior is the pooled rate of the nearest comparables.	0.5726	1.73×	0.02237	0.10920	0.01260
SVR Support-vector regression on the same feature matrix.	0.5699	1.60×	0.02258	0.11313	0.01695
Random forest Forest regression; a learned similarity metric.	0.5932	1.47×	0.02254	0.10986	0.01824
Core (KNN+SVR) Blend of KNN and SVR, no forest.	0.5726	1.73×	0.02237	0.10920	0.01260
Forest blend KNN + SVR + forest, weights fitted out-of-fold.	0.5808	1.60×	0.02240	0.10901	0.01445
Linear regression Ordinary least squares on the same features.	0.5880	1.47×	0.02258	0.11105	0.01856
Logistic regression Binomial logistic on the per-PA home run outcome.	0.5926	1.60×	0.02261	0.11018	0.01912
Neural net (torch) Two-layer PyTorch MLP, PA-weighted, trained on the same features.	0.5864	1.60×	0.02242	0.10922	0.01573
XGBoost Gradient-boosted trees, depth 3 with subsampling and L2.	0.6051	1.87×	0.02273	0.11078	0.02337

Slate agreement

PRIOR	SHARED	ITS TOP SIX
Served (full season)	6/6	Shohei Ohtani 14.6%, Teoscar Hernández 14.2%, Jackson Chourio 13.9%, Will Smith 13.7%, Mookie Betts 13.1%, Freddie Freeman 12.8%
Flat league	6/6	Jackson Chourio 13.8%, Teoscar Hernández 13.5%, Will Smith 13.4%, Shohei Ohtani 13.3%, Mookie Betts 13.0%, Freddie Freeman 12.3%
KNN comps	6/6	Shohei Ohtani 14.9%, Teoscar Hernández 14.4%, Jackson Chourio 14.0%, Will Smith 13.6%, Mookie Betts 12.9%, Freddie Freeman 12.7%
SVR	6/6	Shohei Ohtani 14.6%, Jackson Chourio 14.4%, Teoscar Hernández 14.2%, Will Smith 13.5%, Mookie Betts 13.4%, Freddie Freeman 12.3%
Random forest	6/6	Shohei Ohtani 15.0%, Jackson Chourio 14.6%, Teoscar Hernández 14.4%, Will Smith 13.8%, Mookie Betts 12.9%, Freddie Freeman 12.4%
Core (KNN+SVR)	6/6	Shohei Ohtani 14.9%, Teoscar Hernández 14.4%, Jackson Chourio 14.0%, Will Smith 13.6%, Mookie Betts 12.9%, Freddie Freeman 12.7%
Forest blend	6/6	Shohei Ohtani 14.9%, Teoscar Hernández 14.4%, Jackson Chourio 14.2%, Will Smith 13.7%, Mookie Betts 12.9%, Freddie Freeman 12.6%
Linear regression	6/6	Shohei Ohtani 14.7%, Teoscar Hernández 14.4%, Will Smith 14.3%, Jackson Chourio 14.3%, Mookie Betts 13.2%, Freddie Freeman 12.6%
Logistic regression	6/6	Shohei Ohtani 15.0%, Teoscar Hernández 14.4%, Jackson Chourio 14.3%, Will Smith 14.0%, Mookie Betts 13.2%, Freddie Freeman 12.4%
Neural net (torch)	6/6	Shohei Ohtani 15.0%, Teoscar Hernández 14.5%, Jackson Chourio 14.4%, Will Smith 14.1%, Mookie Betts 13.1%, Freddie Freeman 12.4%
XGBoost	6/6	Shohei Ohtani 15.3%, Jackson Chourio 14.7%, Teoscar Hernández 14.4%, Will Smith 14.0%, Mookie Betts 12.8%, Freddie Freeman 12.3%

Hitter by hitter

The top 40 bats by the served probability.

HITTER	SERVED (FULL SEASON)	FLAT LEAGUE	KNN COMPS	SVR	RANDOM FOREST	CORE (KNN+SVR)	FOREST BLEND	LINEAR REGRESSION	LOGISTIC REGRESSION	NEURAL NET (TO
1 Shohei Ohtani LAD slate	**14.6%**	13.3%	14.9%	14.6%	15.0%	14.9%	14.9%	14.7%	15.0%	
2 Teoscar Hernández LAD	**14.2%**	13.5%	14.4%	14.2%	14.4%	14.4%	14.4%	14.4%	14.4%	
3 Jackson Chourio MIL slate	**13.9%**	13.8%	14.0%	14.4%	14.6%	14.0%	14.2%	14.3%	14.3%	
4 Will Smith LAD	**13.7%**	13.4%	13.6%	13.5%	13.8%	13.6%	13.7%	14.3%	14.0%	
5 Mookie Betts LAD	**13.1%**	13.0%	12.9%	13.4%	12.9%	12.9%	12.9%	13.2%	13.2%	
6 Freddie Freeman LAD	**12.8%**	12.3%	12.7%	12.3%	12.4%	12.7%	12.6%	12.6%	12.4%	
7 William Contreras MIL	**12.1%**	12.0%	12.1%	11.9%	12.0%	12.1%	12.1%	12.0%	12.1%	
8 Jake Bauers MIL	**10.8%**	10.2%	11.3%	11.1%	11.0%	11.3%	11.3%	10.9%	10.9%	
9 Andy Pages LAD	**10.6%**	10.2%	10.5%	10.1%	10.3%	10.5%	10.4%	10.1%	10.0%	
10 Garrett Mitchell MIL	**10.3%**	9.7%	10.0%	9.1%	9.5%	10.0%	9.8%	9.5%	9.4%	
11 Brice Turang MIL	**9.9%**	10.2%	9.8%	9.8%	9.7%	9.8%	9.8%	9.4%	9.3%	
12 Kyle Tucker LAD	**8.7%**	8.9%	8.7%	8.6%	8.7%	8.7%	8.7%	8.6%	8.4%	
13 Christian Yelich MIL	**8.6%**	8.8%	9.0%	8.4%	8.9%	9.0%	9.0%	9.0%	8.5%	
14 Tommy Edman LAD	**6.8%**	7.5%	6.5%	6.7%	6.3%	6.5%	6.4%	6.7%	6.7%	
15 Joey Ortiz MIL	**5.1%**	6.1%	5.2%	5.4%	5.4%	5.2%	5.3%	5.1%	5.2%	
16 Cooper Pratt MIL	**4.9%**	6.5%	4.9%	4.0%	4.7%	4.9%	4.8%	4.3%	4.4%	
17 Sal Frelick MIL	**4.2%**	5.5%	4.5%	4.2%	4.1%	4.5%	4.4%	4.2%	4.2%	

Per-game probabilities: each prior's per-PA rate run through this hitter's own plate appearances, park, weather, bullpen and times-through-the-order terms. Only the prior changes across the columns — every other term is held fixed, so a left-to-right spread is the prior disagreeing about the hitter, not the context. Shading grows with the probability.

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